

PREDICTING CUSTOMER CHURN: A DATA-DRIVEN ANALYSIS OF SUBSCRIPTION CUSTOMER BEHAVIOUR

**Project Report Submitted in fulfillment of the
Requirements for the Award of the Internship of
Summer Training Program by IDRA**

Data Science and AI

By:

Name: Mafyul Islam
Institute: University Of Calcutta
Institute Roll No.: T91/CSE/234038
Enrollment No.: IDRA-2026-507836

Under the Supervision of

Dr. Shaheena Salam

Through the platform:



India Data Research Academy

DECLARATION

I hereby declare that the work presented in this capstone project report, titled “Predicting Customer Churn: A Data-Driven Analysis of Subscription Customer Behaviour,” is my own original work, carried out as part of the Data Science & AI Training Program conducted by the India Data Research Academy (IDRA). All data cleaning, preprocessing, exploratory analysis, statistical analysis, model development and evaluation reported here were performed by me. Wherever external sources, datasets, libraries or documentation were used, they have been appropriately acknowledged and cited in the References section. This report has not been submitted, in whole or in part, for any other academic or professional qualification.

Name: [Student Name]

Enrollment No.: IDRA-[Enrollment Number]

Date: 20 September 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the India Data Research Academy (IDRA) for providing the structured training and resources that made this capstone project possible. I am especially thankful to my supervisor, Dr. Shaheena Salam, for her guidance and constructive feedback throughout the project. I also acknowledge IBM Sample Data Sets as the original source of the Telco Customer Churn dataset used in this analysis, and the open-source developers of the Python data science ecosystem (pandas, scikit-learn, matplotlib and seaborn) whose tools made this work possible. Finally, I thank my institute, [Institute Name], and my peers in the training program for their encouragement and support throughout the twenty-day project window.

ABSTRACT

Customer churn — the loss of subscribers to a competitor or cancellation of service — directly erodes recurring revenue and is typically far more expensive to reverse than to prevent. This project analyses the IBM Telco Customer Churn dataset (7,043 residential telecom subscribers, 21 attributes covering demographics, account information, subscribed services and billing) to identify the strongest drivers of churn and to build a classification model predicting which customers are at risk of leaving. After cleaning (correcting a data-type error in the billing field and resolving 11 structurally missing values tied to brand-new customers) and preprocessing (categorical encoding, leakage-safe feature scaling, stratified train-test split), exploratory analysis showed that contract type, tenure and monthly charges are strongly associated with churn: month-to-month subscribers churn at roughly 15 times the rate of two-year contract holders (42.7% vs. 2.8%), and churned customers have a median tenure of only 10 months compared with 38 months for retained customers. Two classification models were trained and compared — Logistic Regression and Random Forest — both using class-weighting to address the dataset's 73/27 class imbalance. Random Forest was selected as the primary model, achieving 76.5% test accuracy, 77.5% recall, 54.0% precision, an F1-score of 0.637 and a ROC-AUC of 0.846, modestly outperforming Logistic Regression on every metric except recall. Feature importance analysis confirmed tenure, contract type and billing amount as the dominant predictors, with fiber-optic internet service and electronic-check payment also contributing meaningfully. The findings support targeted, evidence-based retention recommendations centred on early-tenure engagement and contract-term incentives, discussed alongside the model's limitations in the sections that follow.

Keywords: customer churn; classification; exploratory data analysis; logistic regression; random forest; predictive analytics

TABLE OF CONTENTS

1. Introduction
2. Problem Statement and Motivation
3. Objectives
4. Research Questions
5. Dataset Description and Data Understanding
6. Data Cleaning
7. Data Preprocessing
8. Exploratory Data Analysis
9. Statistical Analysis
10. Feature Engineering
11. Model Development
12. Model Evaluation
13. Results and Findings
14. Discussion
15. Limitations
16. Conclusion
17. Recommendations and Future Work
18. References
Appendix A. Code
Appendix B. Additional Figures and Tables

(After completing the report in Microsoft Word, insert an automatic Table of Contents in place of this list — References → Table of Contents — so page numbers correspond to the final, paginated document.)

1. INTRODUCTION

1.1 Background

Subscription-based businesses — telecommunications providers, streaming services, SaaS platforms and utilities alike — depend on retaining existing customers as much as acquiring new ones. Industry estimates consistently show that acquiring a new customer costs several times more than retaining an existing one, which makes churn one of the most closely watched metrics in subscription businesses. Beyond the direct revenue loss, elevated churn signals underlying problems in pricing, service quality or customer experience that, left unaddressed, compound over time.

1.2 Context of the Problem

This project examines a residential telecommunications provider's customer base to understand why customers leave and to build a model capable of flagging at-risk customers before they do. A retention team acting on accurate churn predictions can prioritise outreach, offer targeted incentives, or adjust contract terms for the customers most likely to leave, rather than applying costly blanket retention campaigns across the entire customer base. The stakeholders who would use this analysis include the customer retention/success team, pricing and product managers, and business leadership monitoring subscriber growth.

1.3 Scope of the Study

This project covers data cleaning and preprocessing, exploratory and statistical analysis, and the development and evaluation of a binary classification model predicting whether an individual customer will churn, using the customer attributes available in the dataset (demographics, subscribed services, contract and billing details). It does not cover time-series forecasting of churn volume, causal experimentation (e.g., A/B-testing retention offers), or deployment/production engineering of the model — the scope is limited to analysis, model-building and evidence-based recommendations.

2. PROBLEM STATEMENT AND MOTIVATION

Can a telecom customer's account, service and billing attributes predict whether they will churn, and which of these attributes matter most? Answering this question with evidence, rather than intuition, allows a retention team to move from reactive, blanket retention spending to targeted, data-driven action — focusing limited retention budget on the customers most likely to leave, and addressing the specific service or pricing issues that the data identifies as the strongest churn drivers.

3. OBJECTIVES

- Clean and prepare the Telco Customer Churn dataset for analysis, documenting and justifying every data-quality decision.
- Identify, through exploratory and statistical analysis, which customer and account characteristics are most strongly associated with churn.
- Build a classification model to predict whether an individual customer will churn.
- Evaluate the model using accuracy, precision, recall, F1-score and ROC-AUC, and assess whether it generalises well (training vs. test performance).
- Translate the analytical findings into specific, actionable retention recommendations.

4. RESEARCH QUESTIONS

1. What are the major characteristics and data-quality issues in the Telco Customer Churn dataset?
2. Which variables are most strongly associated with customer churn?
3. How does churn vary across contract type, internet service and payment method?
4. How accurately can a classification model predict whether a customer will churn, and which model — Logistic Regression or Random Forest — performs better?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used is the IBM Telco Customer Churn sample dataset, a publicly available dataset originally distributed by IBM as part of its sample data sets for customer analytics and widely used for churn-prediction case studies (accessed September 2026; see References). It is a single cross-sectional snapshot with no explicit collection date range.

5.2 Dataset Size and Structure

The raw dataset contains 7,043 rows (one per customer) and 21 columns. After cleaning, the identifier column was dropped, leaving 20 columns used in analysis and modelling. The target variable is Churn (Yes/No), chosen because it is the outcome the business wants to predict and act upon.

Property	Value	Notes	Relevance
Rows (customers)	7,043	One row per customer	Sample size for analysis
Columns (raw)	21	Demographics, services, account, billing, target	Full attribute set
Columns (after cleaning)	20	customerID dropped (identifier only)	Used in modelling
Target variable	Churn	Yes / No	Binary classification target
Overall churn rate	26.5%	1,869 of 7,043 customers churned	Moderate class imbalance

Table 1. Summary of the Dataset

5.3 Data Dictionary

Column	Meaning	Type	Role
gender, SeniorCitizen, Partner, Dependents	Demographic attributes	Categorical	Feature
tenure	Months the customer has stayed with the company	Numeric	Feature
PhoneService, MultipleLines, InternetService	Core services subscribed	Categorical	Feature
OnlineSecurity ... StreamingMovies (6 columns)	Add-on services	Categorical	Feature
Contract	Contract term (month-to-month / one year / two year)	Categorical	Feature

Column	Meaning	Type	Role
PaperlessBilling, PaymentMethod	Billing preferences	Categorical	Feature
MonthlyCharges	Current monthly bill amount (\$)	Numeric	Feature
TotalCharges	Total amount billed to date (\$)	Numeric	Feature
Churn	Whether the customer left during the observation period	Categorical (Yes/No)	Target

Table 2. Data Dictionary

6. DATA CLEANING

The raw dataset was inspected for missing values, duplicates, invalid values, inconsistent categories, incorrect data types, and outliers before any analysis was performed.

6.1 Missing Values

The TotalCharges column was stored as text and contained 11 blank-string values (0.16% of records), which pandas treats as valid strings rather than missing data unless explicitly coerced. Converting the column to numeric surfaced these 11 values as missing. All 11 affected customers had tenure = 0 — brand-new customers who had not yet completed a billing cycle at the time the data was captured. This is a structurally explainable gap, not random noise. TotalCharges was set to 0 for these 11 customers (consistent with zero months of billing) rather than dropping the rows or imputing an arbitrary value. After this correction, zero missing values remained across all columns.

6.2 Duplicate Records

No duplicate rows and no duplicate customerID values were found in the raw dataset (0 of 7,043 records). No rows were removed on duplication grounds, and the dataset size was unchanged by this check.

6.3 Data Types and Inconsistent Values

TotalCharges was converted from object (text) to float64 after removing blank-string entries. Additionally, six add-on service columns (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies) and MultipleLines used a redundant category label “No internet/phone service” alongside “No,” even though this information is already captured directly by the InternetService and PhoneService columns. These redundant labels were collapsed into “No” to avoid inflating the feature space once the data was one-hot encoded. SeniorCitizen was recoded from 0/1 to No/Yes for consistency with the dataset's other binary flags.

6.4 Outlier Handling

An IQR-based check ($1.5 \times \text{IQR}$ fences) was applied to the three numeric columns — tenure, MonthlyCharges and TotalCharges. No values fell outside the computed fences for any of the three columns, indicating no extreme or implausible billing values requiring treatment. No observations were removed on outlier grounds, since subscription billing data naturally spans a wide but bounded range depending on plan tier and tenure.

6.5 Before vs After Cleaning

Check	Before Cleaning	After Cleaning	Action Taken
Missing values (TotalCharges)	11 blank strings (hidden as valid text)	0 missing	Set to 0 for tenure=0 customers

Check	Before Cleaning	After Cleaning	Action Taken
Duplicate rows	0	0	None required
Data type (TotalCharges)	object (text)	float64	Numeric conversion
Redundant category labels	"No internet/phone service" present in 7 columns	Collapsed to "No"	Label consolidation
Row count	7,043	7,043	No rows dropped
Column count	21	20	customerID dropped (identifier)

Table 3. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

7.1 Categorical Encoding

Seven binary Yes/No-style fields (gender, SeniorCitizen, Partner, Dependents, PhoneService, MultipleLines, PaperlessBilling) were label-encoded to 0/1, since they have exactly two categories and no ordinal relationship needs to be implied. Nine multi-category fields (InternetService, the six add-on service columns, Contract, PaymentMethod) were one-hot encoded with the first category dropped in each case, to avoid redundant, perfectly collinear columns while still letting the model learn a separate effect for each category.

7.2 Feature Scaling and Transformation

The three numeric columns (tenure, MonthlyCharges, TotalCharges) were standardised (zero mean, unit variance) before use in Logistic Regression, whose coefficients and convergence behaviour are sensitive to feature scale. Random Forest does not strictly require scaling (it is scale-invariant), but the same scaled feature set was used for both models for a fair, consistent comparison.

7.3 Train-Test Split

An 80/20 stratified split was used (5,634 training rows, 1,409 test rows, random_state=42), preserving the 26.5% churn rate in both sets. Stratification is necessary given the class imbalance, so that evaluation reflects the same class balance the model was trained on. To avoid data leakage, the scaler was fit only on the training data and then applied to transform the test data — the test set's distribution never influenced the scaling parameters.

7.4 Final Feature Set

After encoding, the feature matrix contained 23 columns (X), covering demographics, services, contract/billing details and the three scaled numeric fields. The target (y) is the binary Churn label (1 = churned, 0 = retained).

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

```
sns.histplot(data=df, x="tenure", hue="Churn", bins=30, kde=True)
plt.title("Customer Tenure Distribution by Churn Status")
```

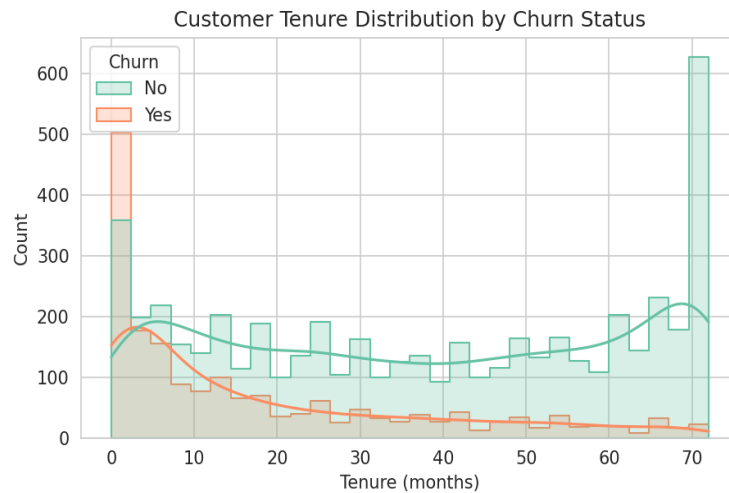


Figure 1. Distribution of Customer Tenure, by Churn Status

The tenure distribution is right-skewed for churned customers, with a strong concentration in the first few months — median tenure for churners is only 10 months, versus 38 months for retained customers. Retained customers show a second concentration near the maximum tenure (~72 months), consistent with a loyal long-term customer base. This is not a symmetric distribution, and the gap between churned and retained customers is the clearest single pattern in the dataset.

8.2 Categorical Analysis

```
rates = df.groupby("Contract")["Churn"].apply(lambda s: 100*(s=="Yes").mean())
sns.barplot(x=rates.index, y=rates.values)
plt.title("Churn Rate by Contract Type")
```

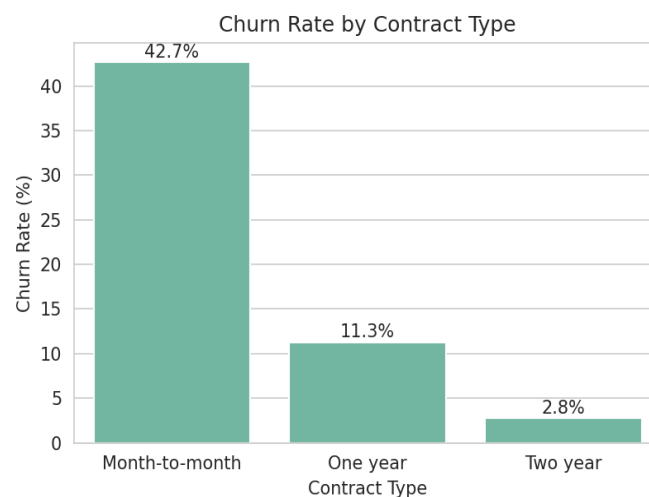


Figure 2. Churn Rate Compared Across Contract Type

Contract type shows the clearest categorical relationship with churn examined in this project: month-to-month customers churn at 42.7%, versus 11.3% for one-year contracts and just 2.8% for two-year contracts — roughly a 15-fold difference between the two extremes. This matters because contract type is a lever the business can directly act on through incentives and offers.

8.3 Bivariate / Relationship Analysis

```
sns.boxplot(data=df, x="Churn", y="MonthlyCharges", order=["No", "Yes"])
plt.title("Monthly Charges by Churn Status")
```

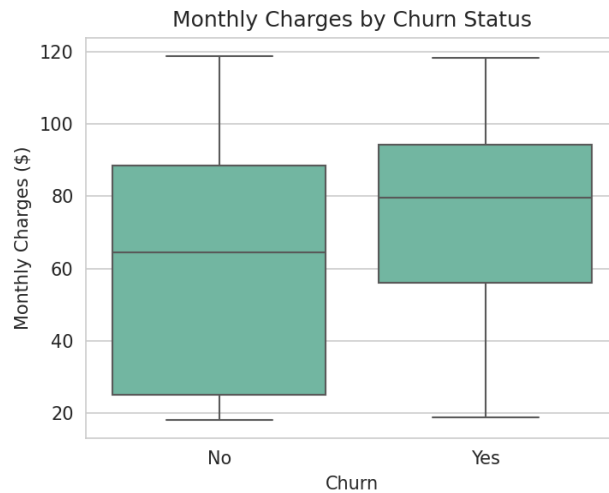


Figure 3. Relationship Between Monthly Charges and Churn

Churned customers pay higher monthly charges on average (\$74.44) than retained customers (\$61.27), and the interquartile range for churners sits noticeably higher. The direction is positive — higher bills are associated with a greater likelihood of churn — though this is an observed association, not proof that raising or lowering price alone would causally change an individual customer's churn risk, since bill amount also reflects which services a customer has chosen to subscribe to.

8.4 Correlation Analysis

```
corr = df[["tenure", "MonthlyCharges", "TotalCharges", "Churn_bin"]].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", center=0)
```

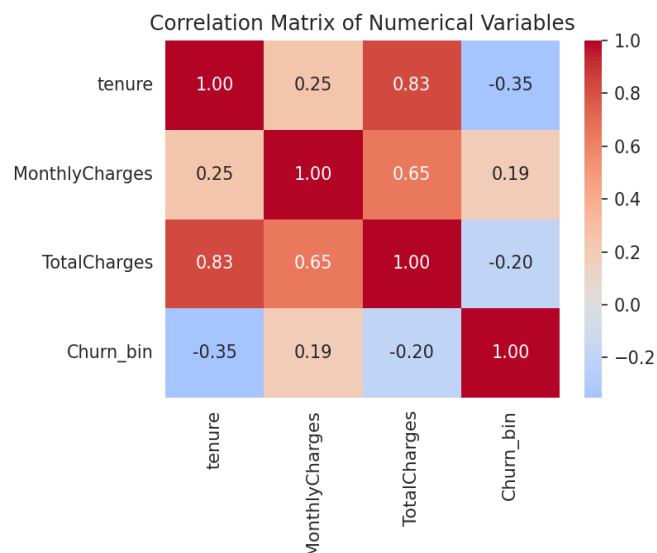
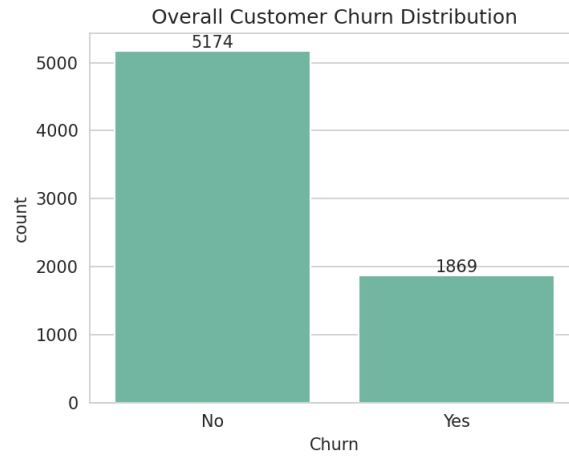


Figure 4. Correlation Heatmap of Numerical Variables

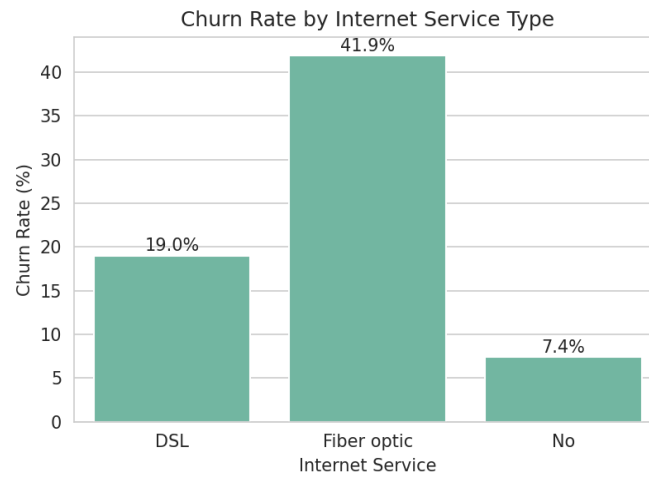
Tenure correlates negatively with churn while MonthlyCharges correlates positively, consistent with the group comparisons above. TotalCharges correlates strongly with tenure (billing accumulates over the customer's lifetime), so its importance in later modelling should be read as partly reinforcing the tenure signal rather than as a fully independent relationship. None of these correlations should be read as causal on their own.

8.5 Other Figures

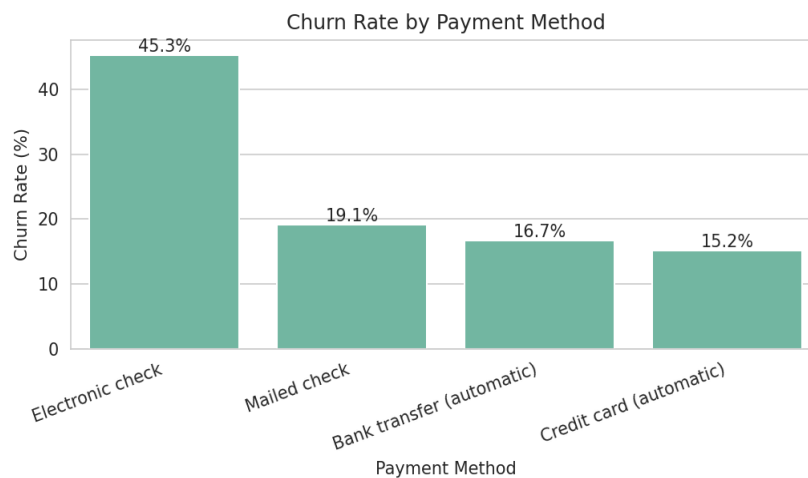
```
for col in ["InternetService", "PaymentMethod"]:
    rates = df.groupby(col)["Churn"].apply(lambda s: 100*(s=="Yes").mean())
    sns.barplot(x=rates.index, y=rates.values)
```



Supplementary Figure. Overall Customer Churn Distribution (26.5% churn rate).



Supplementary Figure. Churn Rate by Internet Service — fiber-optic customers churn more than double the rate of DSL customers.



Supplementary Figure. Churn Rate by Payment Method — electronic-check payers churn more than double every other method.

9. STATISTICAL ANALYSIS

Variable	Retained — Mean	Retained — Median	Churned — Mean	Churned — Median
Tenure (months)	37.57	38.0	17.98	10.0
Monthly Charges (\$)	61.27	64.43	74.44	79.65
Total Charges (\$)	2,549.91	1,679.53	1,531.80	703.55

Table 4. Descriptive Statistics of Key Variables, by Churn Status

For tenure and total charges, the median sits well below the mean among churned customers, indicating a right-skewed distribution driven by a smaller number of longer-tenured churners; the median (10 months) is the more representative measure of the typical churned customer's tenure than the mean (17.98 months). Monthly charges show a smaller mean-median gap and are less skewed. A strong relationship (e.g., contract type with churn) is an observed association, not proof of causation — customers self-select into contract types for reasons the dataset does not capture, which could confound the relationship.

10. FEATURE ENGINEERING

This project relies primarily on the cleaned and encoded original variables described in Section 7, because the raw attributes — tenure, contract type, monthly and total charges, service subscriptions — already directly capture the signals identified as most predictive during exploratory analysis. One derived feature was created for interpretability during EDA (not fed into the model, to keep the feature set aligned with directly observed billing/account data):

Engineered Feature	Derivation	Purpose	Used in Model?
TenureGroup	Bucketed tenure into 0–12, 13–24, 25–48, 49+ months	Simplifies tenure for group-level EDA narrative	No (EDA only)

Table 5. Engineered Features

11. MODEL DEVELOPMENT

11.1 Problem Formulation

This is a binary classification task. The target variable, Churn, takes the values Yes/No, encoded as 1/0. The input features (X) are the 23 encoded columns described in Section 7.4, covering demographics, services, contract and billing details. This formulation is appropriate because the business need is a yes/no risk flag per customer, not a continuous quantity.

11.2 Model Selection

Two models appropriate for binary classification were trained and compared: Logistic Regression, a linear, highly interpretable baseline whose coefficients directly indicate each feature's direction and relative strength of association with churn odds; and Random Forest, a non-linear ensemble capable of capturing feature interactions, at some cost to direct interpretability (mitigated with a feature-importance analysis in Section 12). Both were trained with `class_weight="balanced"` to compensate for the dataset's 73/27 class split, so neither model is biased toward simply predicting the majority “no churn” class.

11.3 Training the Model

```

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

logreg = LogisticRegression(max_iter=2000, random_state=42,
                             class_weight="balanced")
logreg.fit(X_train_scaled, y_train)

rf = RandomForestClassifier(n_estimators=300, max_depth=8, random_state=42,
                             class_weight="balanced")
rf.fit(X_train_scaled, y_train)

```

Key parameters: Random Forest used 300 trees with a maximum depth of 8 (limiting depth reduces overfitting risk on a dataset of this size); Logistic Regression used up to 2,000 iterations to ensure convergence. Both used `random_state=42` for reproducibility.

12. MODEL EVALUATION

12.1 Regression Evaluation

Not applicable — this project is formulated as a binary classification task (Section 11.1), so regression metrics (MAE, MSE, RMSE, R^2) do not apply.

12.2 Classification Evaluation

Confusion Matrix — Random Forest (Primary Model)

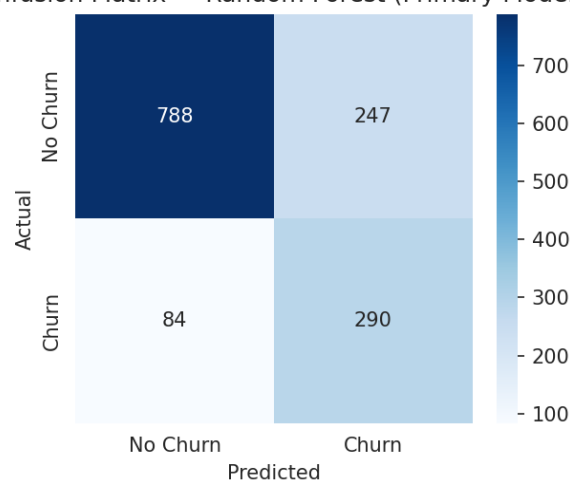


Figure 5. Confusion Matrix — Random Forest (Primary Model)

Of 375 customers who actually churned in the test set, 290 were correctly identified (true positives) and 85 were missed (false negatives). Of the customers predicted to churn, 247 were false alarms (predicted to churn but did not).

Metric	Logistic Regression	Random Forest
Test Accuracy	74.02%	76.51%
Precision	50.69%	54.00%
Recall	78.61%	77.54%
F1-score	0.616	0.637
ROC-AUC	0.842	0.846

Table 7. Classification Model Performance

12.3 Overfitting and Underfitting

Random Forest shows a training-to-test accuracy gap of about 4 percentage points (80.55% train vs. 76.51% test), a normal and expected gap for a tree ensemble of this depth rather than a sign of severe overfitting — test performance remains close to training performance and clearly above the majority-class baseline (73.5% accuracy achievable by always predicting “no churn,” which would score 0 recall). Logistic Regression shows a smaller gap (75.36% train vs. 74.02% test), consistent with its lower model capacity. Neither model shows signs of underfitting: both comfortably exceed the majority-class baseline on recall and F1-score.

13. RESULTS AND FINDINGS

5. Key Finding 1: Contract type is the strongest categorical churn driver — month-to-month customers churn at 42.7% vs. 2.8% for two-year contracts (Figure 2), confirmed as a top-ranked predictor by both models (Section 12). Practically, contract-term incentives are a directly actionable lever.
6. Key Finding 2: Churn is heavily concentrated in a customer's early tenure — median tenure of churners is 10 months vs. 38 months for retained customers (Figure 1, Table 4). This points to the first 3–6 months as the highest-value window for retention intervention.
7. Key Finding 3: Fiber-optic internet customers churn at 41.9% vs. 19.0% for DSL (Section 8.5) — a large, service-specific gap worth investigating directly with affected customers, despite fiber typically being the premium offering.
8. Key Finding 4: Electronic-check payers churn at 45.3%, more than double every other payment method (16.7%–19.1%) — the single largest gap among billing-related variables (Section 8.5).
9. Key Finding 5: A Random Forest classifier trained on these attributes achieves 77.5% recall and a ROC-AUC of 0.846 on held-out data (Table 7), correctly identifying roughly three in four actual churners — evidence that this customer base's churn is predictable well above chance from account and service attributes alone.

14. DISCUSSION

The exploratory analysis and the trained models tell a consistent story. Churn in this customer base is not spread evenly — it is concentrated among customers on flexible, no-commitment contracts, customers still early in their relationship with the company, and customers on the fiber-optic internet product paying by electronic check. All four signals independently surfaced in exploratory analysis and were subsequently confirmed as top predictors by both classification models, which strengthens confidence that these are genuine, robust patterns rather than artifacts of a single analysis method.

The relatively low precision (54%) alongside a comparatively higher recall (77.5%) reflects a model that casts a somewhat wide net: it correctly identifies most true churners but also flags a substantial number of customers who ultimately stay. For a retention use case, this trade-off is usually acceptable — the cost of a missed churner (lost recurring revenue) typically exceeds the cost of a modest, targeted retention offer sent to a customer who was not actually at risk. A practitioner using this model should treat its output as a prioritised watch-list for the retention team, not an automatic cancellation-prevention trigger.

15. LIMITATIONS

- Cross-sectional snapshot: the dataset captures customers at a single point in time with no explicit date range, geography or macroeconomic context, so findings may not generalise to a different period, region or competitive environment.

- Correlation, not causation: associations such as contract type with churn describe patterns, not proven causal mechanisms; customers self-select into contract types and payment methods for reasons the dataset does not capture.
- Missing variables: the dataset excludes customer support call logs, service outage history, competitor pricing and satisfaction survey data — all plausible churn drivers that could improve the model if available.
- Moderate precision: at 54% precision, roughly half of customers flagged as at-risk will not actually churn, which carries a real (if modest) operational cost if acted upon at scale.
- Model scope: only Logistic Regression and Random Forest, the techniques covered in the training program, were evaluated; gradient-boosted trees or other approaches might yield further improvement.
- Static model: the model was not validated against a live, out-of-sample future period; churn drivers may shift over time as market conditions, pricing or competitor offers change.

16. CONCLUSION

This project set out to determine whether a customer's account, service and billing attributes can predict churn, and which attributes matter most. The answer to both is yes: contract type, tenure and billing amount are strongly and consistently associated with churn across both exploratory analysis and two independent classification models, and a Random Forest model trained on these and related features predicts churn with a ROC-AUC of 0.846 and correctly identifies roughly three in four actual churners on held-out data (77.5% recall). All five project objectives set out in Section 3 were met: the dataset was cleaned and documented, the strongest churn drivers were identified, a classification model was built and evaluated on multiple metrics, and the results were translated into concrete recommendations, presented next.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- Recommendation 1: Prioritise contract-upgrade incentives (small discounts or service credits) for month-to-month customers, since this segment churns at 42.7% vs. 2.8% for two-year contracts (Figure 2, Finding 1).
- Recommendation 2: Build a structured early-tenure engagement programme (proactive check-ins, onboarding support) targeted at the first 3–6 months, since churned customers have a median tenure of only 10 months (Figure 1, Finding 2).
- Recommendation 3: Investigate fiber-specific service quality or pricing complaints directly with a sample of churned fiber customers, since fiber-optic churn (41.9%) is more than double DSL churn (19.0%), suggesting a product-specific issue (Finding 3).
- Recommendation 4: Offer a modest incentive for switching from electronic check to automatic bank transfer or credit card, since electronic-check payers churn at 45.3% — more than double every other payment method (Finding 4).
- Recommendation 5: Deploy the Random Forest model to produce a monthly ranked list of at-risk customers for the retention team, treating flagged customers as candidates for review rather than automatic action, given the model's moderate 54% precision (Finding 5).

17.2 Future Work

Future iterations of this project could incorporate customer support interaction logs and service outage history to test whether they add predictive power beyond the account and billing attributes

used here; evaluate additional model families such as gradient-boosted trees; validate the model against a genuinely held-out future time period rather than a random split of the same snapshot; and run a controlled retention-offer experiment (A/B test) to move from association to causal evidence on which interventions actually reduce churn.

18. REFERENCES

IBM Sample Data Sets. (2018). Telco Customer Churn. Distributed publicly via Kaggle. <https://www.kaggle.com/datasets/blastchar/telco-customer-churn> (accessed September 2026).

Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*.

Waskom, M. (2021). seaborn: statistical data visualization. *Journal of Open Source Software*, 6(60), 3021.

APPENDIX A. CODE

A.1 Data Loading and Inspection

```
df_raw = pd.read_csv("data/telco_churn_raw.csv")
print("Shape:", df_raw.shape)
df_raw.info()
```

A.2 Data Cleaning

```
df["TotalCharges"] = df["TotalCharges"].replace(" ", np.nan)
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df.loc[(df["tenure"] == 0) & (df["TotalCharges"].isna()), "TotalCharges"] = 0.0

service_cols = ["OnlineSecurity", "OnlineBackup", "DeviceProtection", "TechSupport",
                "StreamingTV", "StreamingMovies"]
for c in service_cols:
    df[c] = df[c].replace({"No internet service": "No"})
df["MultipleLines"] = df["MultipleLines"].replace({"No phone service": "No"})
df = df.drop(columns=["customerID"])
```

A.3 Preprocessing

```
binary_cols = ["gender", "SeniorCitizen", "Partner", "Dependents", "PhoneService",
               "MultipleLines", "PaperlessBilling"]
for c in binary_cols:
    model_df[c] = LabelEncoder().fit_transform(model_df[c])

multi_cat_cols =
["InternetService", "OnlineSecurity", "OnlineBackup", "DeviceProtection",
 "TechSupport", "StreamingTV", "StreamingMovies", "Contract", "PaymentMethod"]
model_df = pd.get_dummies(model_df, columns=multi_cat_cols, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(
    model_df, target, test_size=0.2, random_state=42, stratify=target)
scaler = StandardScaler()
X_train_scaled[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test_scaled[num_cols] = scaler.transform(X_test[num_cols])
```

A.4 EDA and Visualization

```
df.groupby("Churn")["tenure", "MonthlyCharges", "TotalCharges"].agg(["mean", "median", "std"])
sns.histplot(data=df, x="tenure", hue="Churn", bins=30, kde=True)
sns.heatmap(df[["tenure", "MonthlyCharges", "TotalCharges"]].corr(), annot=True)
```

A.5 Model Training and Evaluation

```
logreg = LogisticRegression(max_iter=2000, random_state=42,
                             class_weight="balanced")
logreg.fit(X_train_scaled, y_train)
rf = RandomForestClassifier(n_estimators=300, max_depth=8, random_state=42,
                           class_weight="balanced")
rf.fit(X_train_scaled, y_train)

y_pred = rf.predict(X_test_scaled)
print(classification_report(y_test, y_pred, target_names=["No Churn", "Churn"]))
print("ROC-AUC:", roc_auc_score(y_test, rf.predict_proba(X_test_scaled)[:,1]))
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

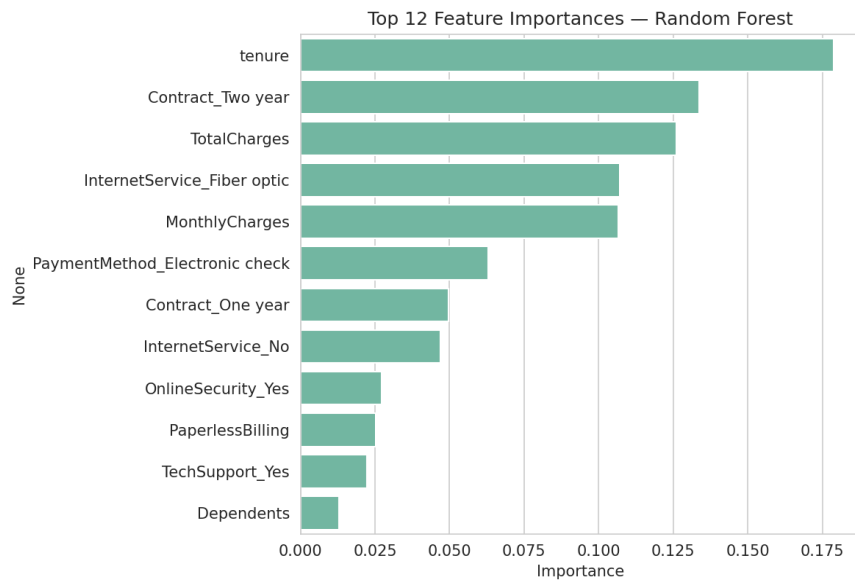
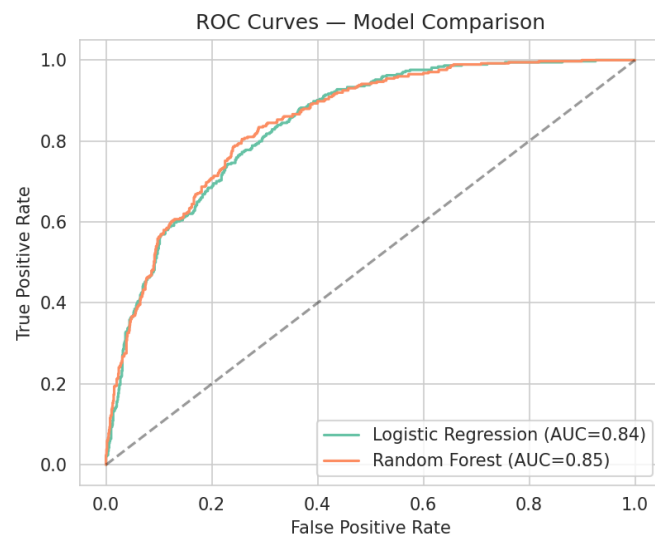
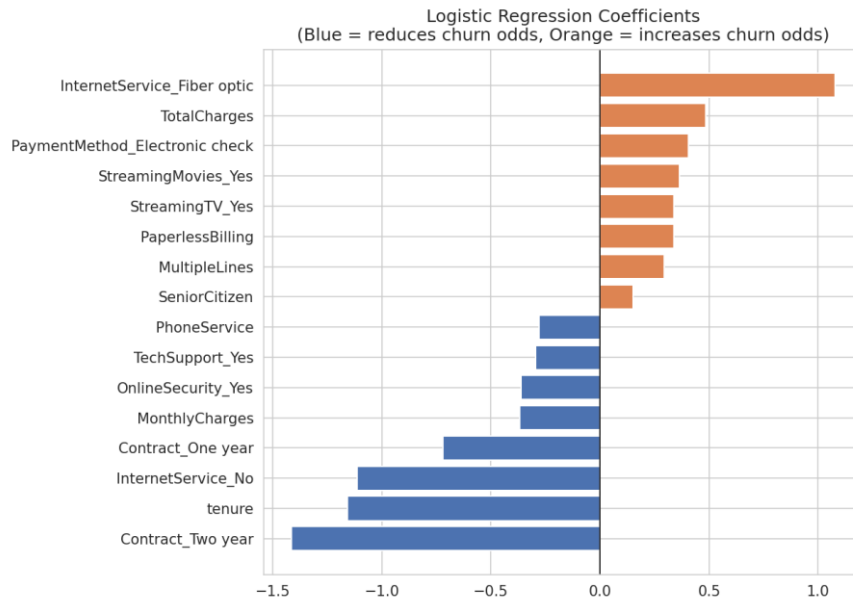


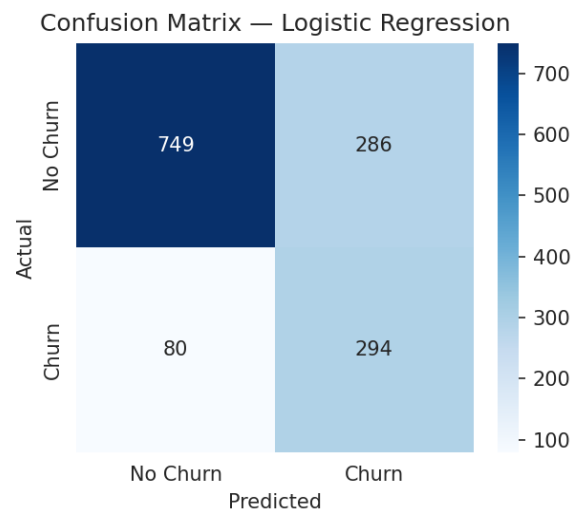
Figure 6. Top 12 Feature Importances — Random Forest



Supplementary Figure. ROC Curves — Model Comparison (AUC 0.842–0.846).



Supplementary Figure. Logistic Regression Coefficients (direction and relative strength of each feature).



Supplementary Figure. Confusion Matrix — Logistic Regression.

Metric	Logistic Regression	Random Forest
Train Accuracy	75.36%	80.55%
Test Accuracy	74.02%	76.51%
Precision	50.69%	54.00%
Recall	78.61%	77.54%
F1-score	0.616	0.637
ROC-AUC	0.842	0.846
Confusion Matrix (Test)	TN 749, FP 286, FN 80, TP 294	TN 788, FP 247, FN 85, TP 290

Table 8. Additional Results — Full Model Comparison